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Stanford, California, USA

Education

2025– **Ph.D.**, Mathematics, Stanford University

2022–2025 **B.Sc.** Mathematics, University College London

- First Class, Ranked 1st out of 341

Research Experience

2026– Quasi Riemann Hypothesis (Supervisor: Dr. Kannan Soundararajan)

- Analyzing the OpenAI breakthrough on the Riemann hypothesis and exploring its applications to L-functions and character sums.

2025– Quantum Chaos and Number Theory (Supervisor: Dr. Kannan Soundararajan)

- Exploring applications of random matrix theory to the statistics of zeros and values of L-functions
- Established a correspondence between Hooley's conjecture on variance of primes in arithmetic progressions and random matrix predictions for L-zeros

2025– Higher Order Fourier Analysis (Supervisor: Dr. Sarah Peluse)

- Exploring Gowers norm and its applications to the theorems of Szemerédi, Green–Tao, and Green–Sawhney

2024–2025 Explicit Quadratic Large Sieve Inequality, (Supervisor: Dr. Ian Petrow)

- Funded by LMS Undergraduate Research Bursaries (URB-2024-41)
- Developed a refined version of Heath-Brown's quadratic large sieve inequality and published the result

2024 Special Values of the Riemann Zeta Function (Supervisor: Dr. Cecilia Busuioc)

- Investigated Bernoulli numbers and computed $\zeta(2k)$ for integers $k \geq 1$
- Studied Beukers's proof that $\zeta(3)$ is irrational

2023–2024 p-adic Numbers (Supervisor: Dr. Cecilia Busuioc)

- Joint with Hantang Guo, May Jiang, and Qing Su
- Explored various aspects of p-adic norms and p-adic power series
- Developed a visualization program for Hensel lifting in Python

2023 Axiom of Choice (Supervisor: Dr. Cecilia Busuioc)

- Joint with Yi Liu, Daya Singh, and Tairan Wang
- Investigated equivalent forms, consequences, and the independence of the Axiom of Choice from ZF set theory

2020 PM2.5 Density Prediction

- Studied PM2.5 density using signal processing, Pandas, and LightGBM
- Innovated and published a paper on a new PM2.5 prediction model

2017–2019 Operating System Development

- Created two 32-bit operating systems from scratch
- Implemented keyboard driver and virtual memory using C and x86 assembly

Publications

1. Liu, Z. *Pair correlation of L-zeros and Hooley's conjecture* Available upon request. 2026.
2. Liu, Z. Explicit Quadratic Large Sieve Inequality. *Acta Arithmetica* **223,** 227–252 (2026).
3. Liu, Z. *A Corrected Simplified Proof of Chen's Theorem* arXiv:2203.07871 [math]. Mar. 2022.
4. Liu, Z. *PM2.5 Density Prediction Based on a Two-Stage Rolling Forecast Model Using LightGBM* in *Computing and Data Science* (eds Cao, W., Ozcan, A., Xie, H. & Guan, B.) (Springer Nature Singapore, Singapore, 2021), 228–248. ISBN: 978-981-16-8885-0.

Presentations

Stanford Kiddie Colloquium

1. *Binary quadratic forms and the zeros of Epstein's zeta function.* June 2026.

Student-Run Seminars on Number Theory and Function Theory

1. *On the distribution of zeros of $\zeta(s)$.* July 2025.
2. *Ω results in number theory.* July 2023.
3. *GPY sieve - from Selberg to Maynard.* July 2022.

UCL Undergraduate Math Colloquium

1. *Special values of the Riemann zeta function.* Oct. 2024.
2. *Introduction to p-adic analysis.* Feb. 2024.
3. *Ray reflection and rational approximation.* Oct. 2023.
4. *Ellipses, pendulum, and double periodicity.* Mar. 2023.
5. *Bounded gaps between primes.* Oct. 2022.

UCL Undergraduate Project Talks

1. *Inverse Galois Problem.* Dec. 2024.
2. *Axiom of Choice: Equivalents, Consequences, and Independence.* June 2023.

Teaching

Stanford University

Teaching Assistantships

2027	Math 120 Groups and Rings (Spring, scheduled)
2027	Math 52 Integral Calculus of Several Variables (Winter, scheduled)
2025–2026	Math 104 Applied Matrix Theory (Autumn, Spring)

Directed Reading Programs

2026	Aredhel Martin	Asymptotic theory of partitions
2026	Karthik Seetharaman	Topics in binary quadratic forms
2026	Robert Dragomirescu	p-adic numbers

Attended Seminars

2026	Qiuzhen International Mathematics Conference (Beijing, China)
2025	Summer School on Analytic Number Theory (Xi'an, China)
2024	International Number Theory Conference in Commemoration of Chengdong Pan (Jinan, China)
2022–2025	London Number Theory Seminars
2022–2025	London Junior Number Theory Seminars
2022–2025	London Heilbronn Colloquia

Other Experience

2020– Math Stack Exchange and Math Overflow

- Answered 250+ questions on complex analysis and number theory
- Earned 8k+ score MSE and 1.4k+ scores on MO as of 2025
- Top answerer under the tag "analytic number theory"

2020– Math Blogging on Zhihu

- Composed 170+ articles for 10 self-initiated math columns
- Answered 980+ math questions from netizens
- Obtained 85k+ followers, 93k+ upvotes, and 81k+ bookmarks to date

2017– Open Source Contributions

- Contributed code to Git, OI Wiki, hello-world, LabCoat, and Manim-Tutorial
- Developed a CSS/JavaScript workaround to address overlapping problem in KaTeX equation rendering.

Awards & Honors

2026	The Halsey L. Royden, Jr. Fellowship
2025–2026	IMA Membership
2025	UCL Dean's List Nomination
2025	Wynne Roberts Prize (Best graduating math student)
2024	LMS Undergraduate Research Bursaries (URB-2024-41)
2024	MSci Prize in Mathematics (Best Year 2 overall performance)
2024	Kestelman Prize (Best Year 2 performance in Real and Complex Analysis)
2023	Filon Prize (Best Year 1 overall performance)
2022	Platinum Division, USA Computing Olympiad
2021	Bronze, S.-T. Yau High School Science Award (Mathematics)
2019	Finalist Award, High School Mathematical Contest in Modeling

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